Evaluation Report — Normal vs Faulty Episodes

H5: /Users/Phips1900/PhD/Research/RelAIBotiX/datasets/IL/smolVLA_base/eval_smolvlabase_mj2_faults_.h5

Episodes: 1000

Success tolerance (XY max-abs): 0.02 m; Target: (0.2, -0.15)

Faulty flags: noise_strength, blur_strength, brightness_strength (episode max > 0)

Summary — success rates & episode stats

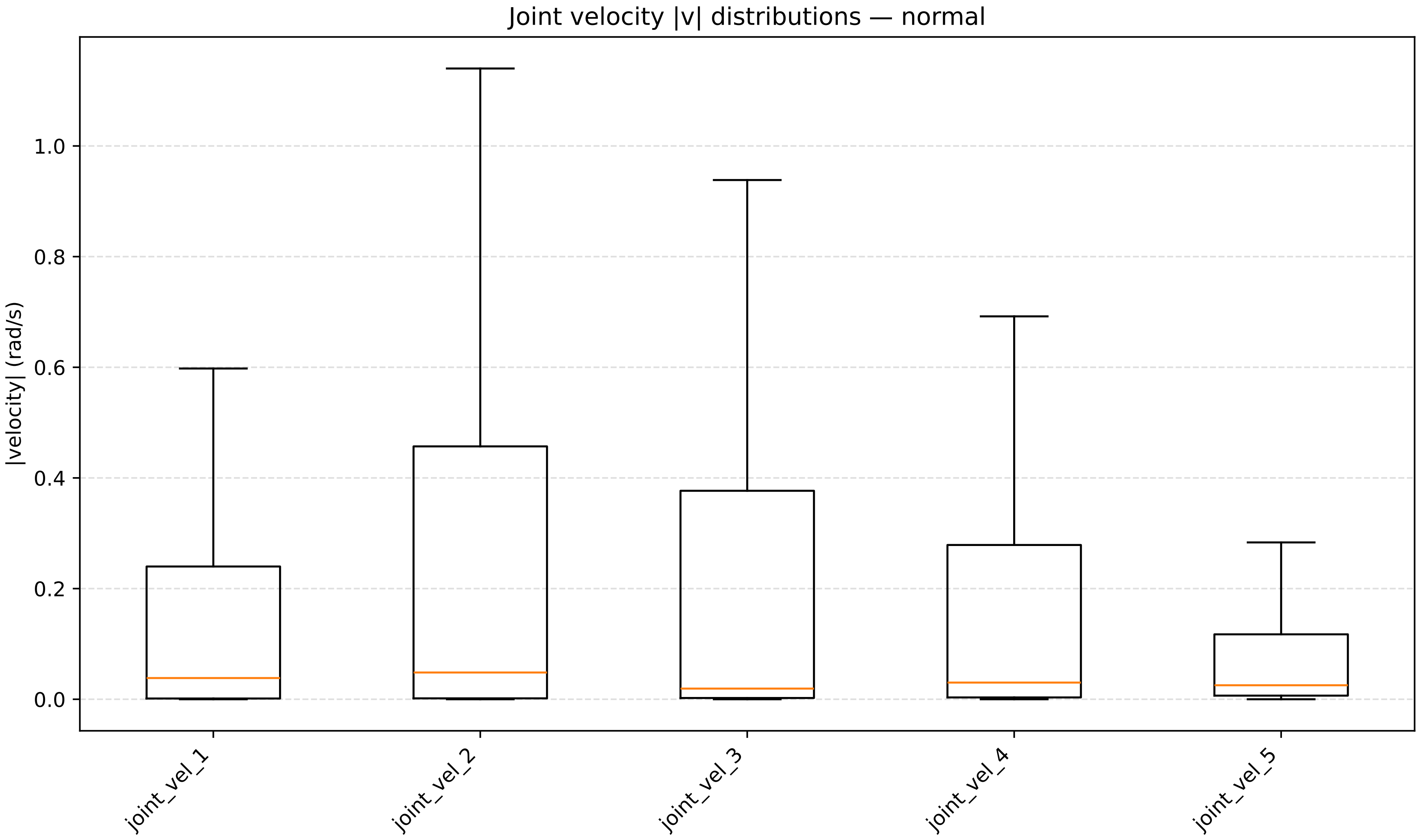
subset	n_episodes	success_rate	mean_pos_err	median_pos_err	avg_duration_sec	avg_n_samples
blur_only	293	40.96	0.109758	0.073717	9.992901	1000.290102
brightness_only	306	76.14	0.042291	0.011328	9.992778	1000.277778
combined	0	NaN	NaN	NaN	NaN	NaN
faulty_any	900	44.33	0.094850	0.048880	9.992667	1000.266667
noise_only	301	15.28	0.133771	0.119533	9.992326	1000.232558
normal	100	96.00	0.013247	0.008542	9.996400	1000.640000

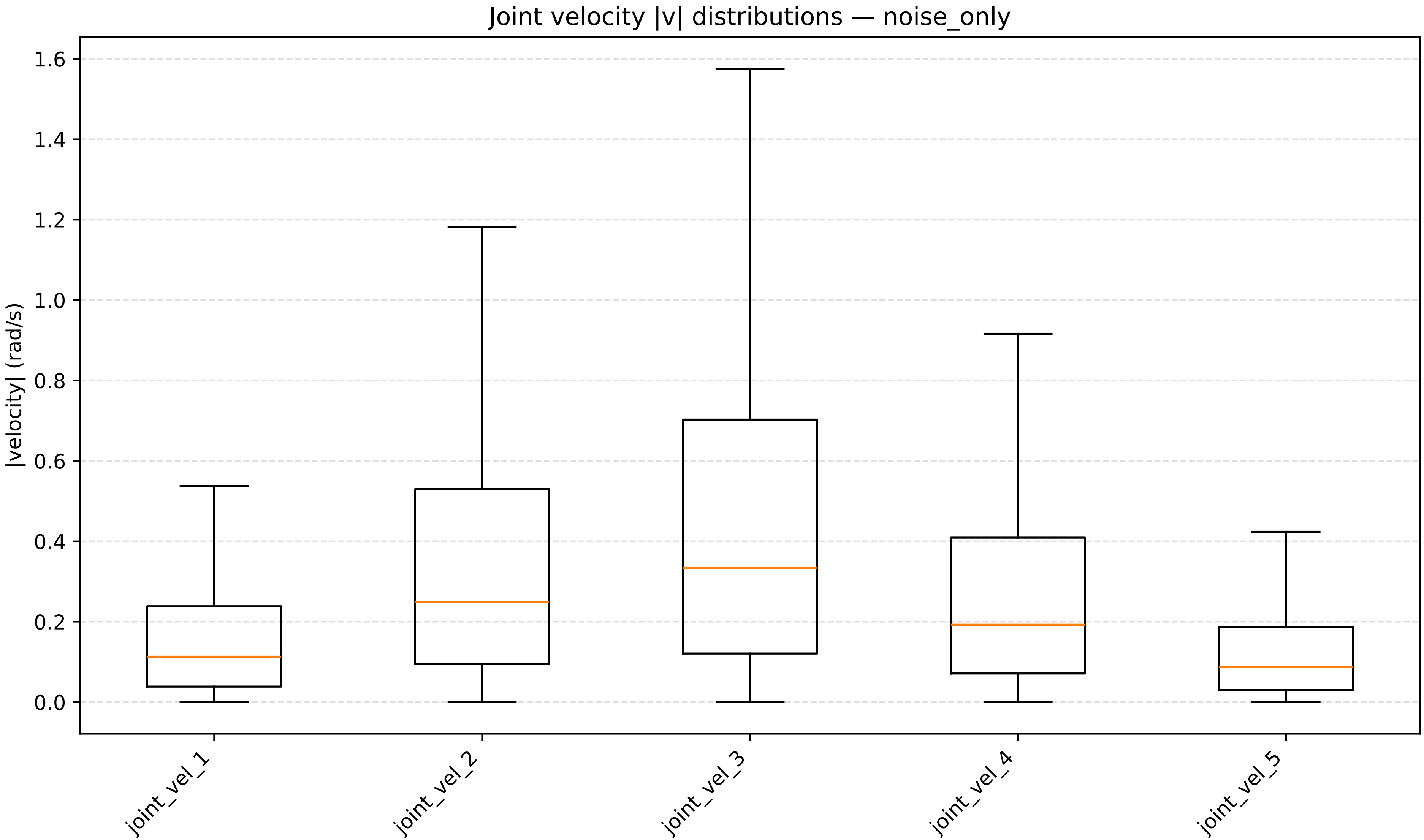
Per-episode (HEAD)

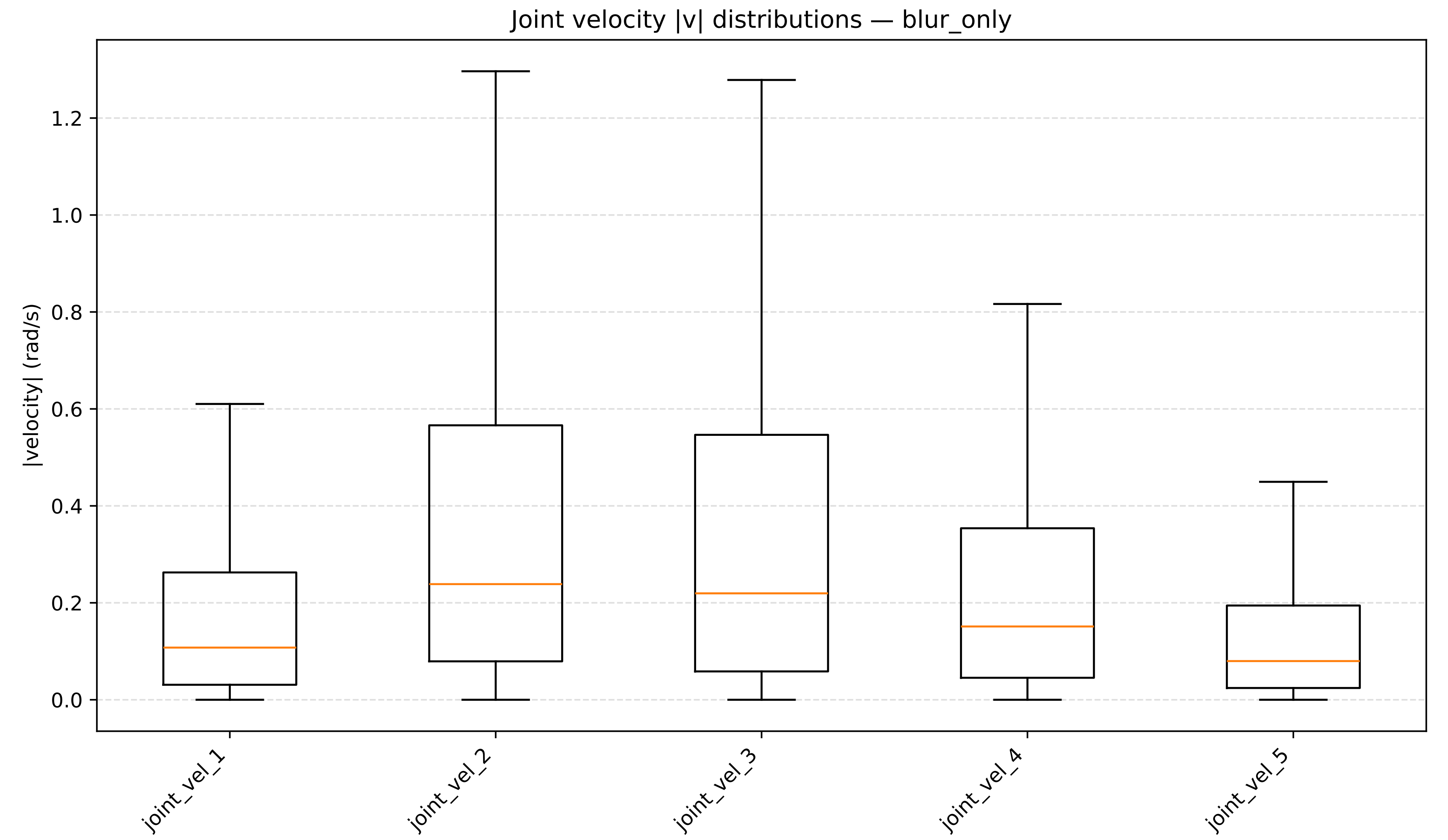
run_id	start_idx	end_idx	n_samples	duration_sec	success	pos_err	goal_x	goal_y	final_x	final_y	noise_strength_max	blur_strength_ma
0	0	999	1000	9.99	True	0.001850	0.2	-0.15	0.201850	-0.151597	0.0	0.
1	1000	2000	1001	10.00	True	0.002746	0.2	-0.15	0.202746	-0.151222	0.0	0.
2	2001	3000	1000	9.99	True	0.009618	0.2	-0.15	0.209618	-0.152493	0.0	0.
3	3001	4000	1000	9.99	True	0.002454	0.2	-0.15	0.200609	-0.147546	0.0	0.
4	4001	5000	1000	9.99	True	0.015292	0.2	-0.15	0.215292	-0.143575	0.0	0.
5	5001	6001	1001	10.00	True	0.014984	0.2	-0.15	0.208966	-0.135016	0.0	0.
6	6002	7002	1001	10.00	True	0.008664	0.2	-0.15	0.197878	-0.158664	0.0	0.
7	7003	8003	1001	10.00	True	0.006625	0.2	-0.15	0.206625	-0.150675	0.0	0.
8	8004	9004	1001	10.00	True	0.010764	0.2	-0.15	0.210764	-0.158594	0.0	0.
9	9005	10005	1001	10.00	True	0.008845	0.2	-0.15	0.208845	-0.154245	0.0	0.
10	10006	11005	1000	9.99	True	0.010390	0.2	-0.15	0.203393	-0.160390	0.0	0.
11	11006	12005	1000	9.99	True	0.010861	0.2	-0.15	0.210861	-0.154189	0.0	0.
12	12006	13005	1000	9.99	True	0.007477	0.2	-0.15	0.206587	-0.157477	0.0	0.
13	13006	14005	1000	9.99	True	0.001833	0.2	-0.15	0.201453	-0.151833	0.0	0.
14	14006	15005	1000	9.99	True	0.012683	0.2	-0.15	0.212683	-0.152455	0.0	0.
15	15006	16005	1000	9.99	True	0.011516	0.2	-0.15	0.211516	-0.148018	0.0	0.
16	16006	17005	1000	9.99	True	0.008674	0.2	-0.15	0.208674	-0.148862	0.0	0.
17	17006	18005	1000	9.99	True	0.006107	0.2	-0.15	0.206107	-0.146857	0.0	0.
18	18006	19005	1000	9.99	True	0.000842	0.2	-0.15	0.199642	-0.149158	0.0	0.
19	19006	20005	1000	9.99	True	0.014636	0.2	-0.15	0.214636	-0.156456	0.0	0.
20	20006	21005	1000	9.99	True	0.004531	0.2	-0.15	0.197348	-0.154531	0.0	0.
21	21006	22005	1000	9.99	True	0.013330	0.2	-0.15	0.203693	-0.163330	0.0	0.
22	22006	23005	1000	9.99	True	0.010240	0.2	-0.15	0.206197	-0.160240	0.0	0.
23	23006	24005	1000	9.99	True	0.007851	0.2	-0.15	0.201291	-0.142149	0.0	0.
24	24006	25005	1000	9.99	True	0.006403	0.2	-0.15	0.203660	-0.156403	0.0	0.
25	25006	26005	1000	9.99	True	0.010838	0.2	-0.15	0.210668	-0.160838	0.0	0.
26	26006	27005	1000	9.99	True	0.008250	0.2	-0.15	0.208250	-0.153759	0.0	0.
27	27006	28005	1000	9.99	True	0.019886	0.2	-0.15	0.219886	-0.133793	0.0	0.
28	28006	29005	1000	9.99	True	0.005857	0.2	-0.15	0.203905	-0.144143	0.0	0.
29	29006	30005	1000	9.99	True	0.012820	0.2	-0.15	0.210936	-0.162820	0.0	0.

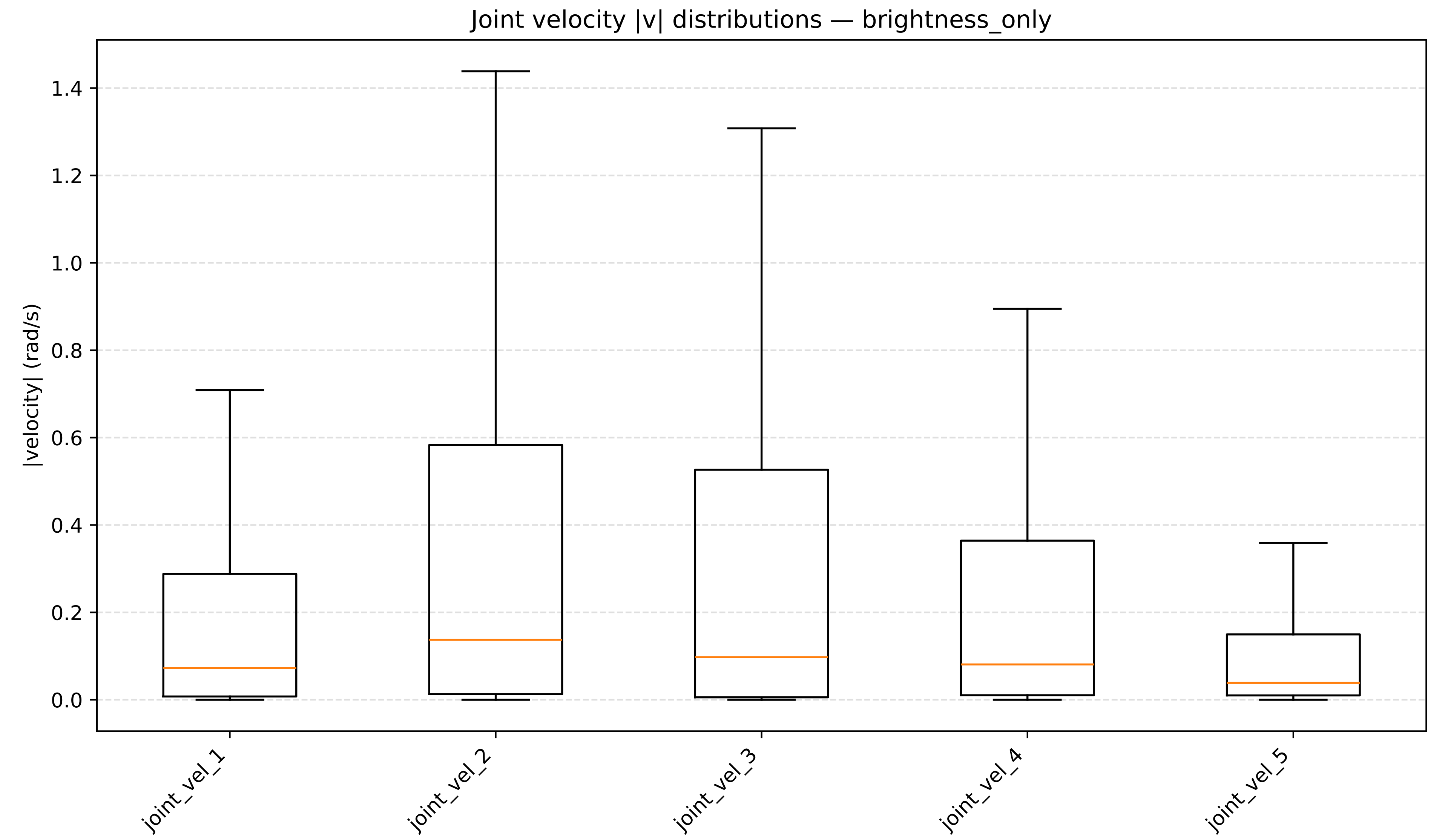
Per-episode (TAIL)

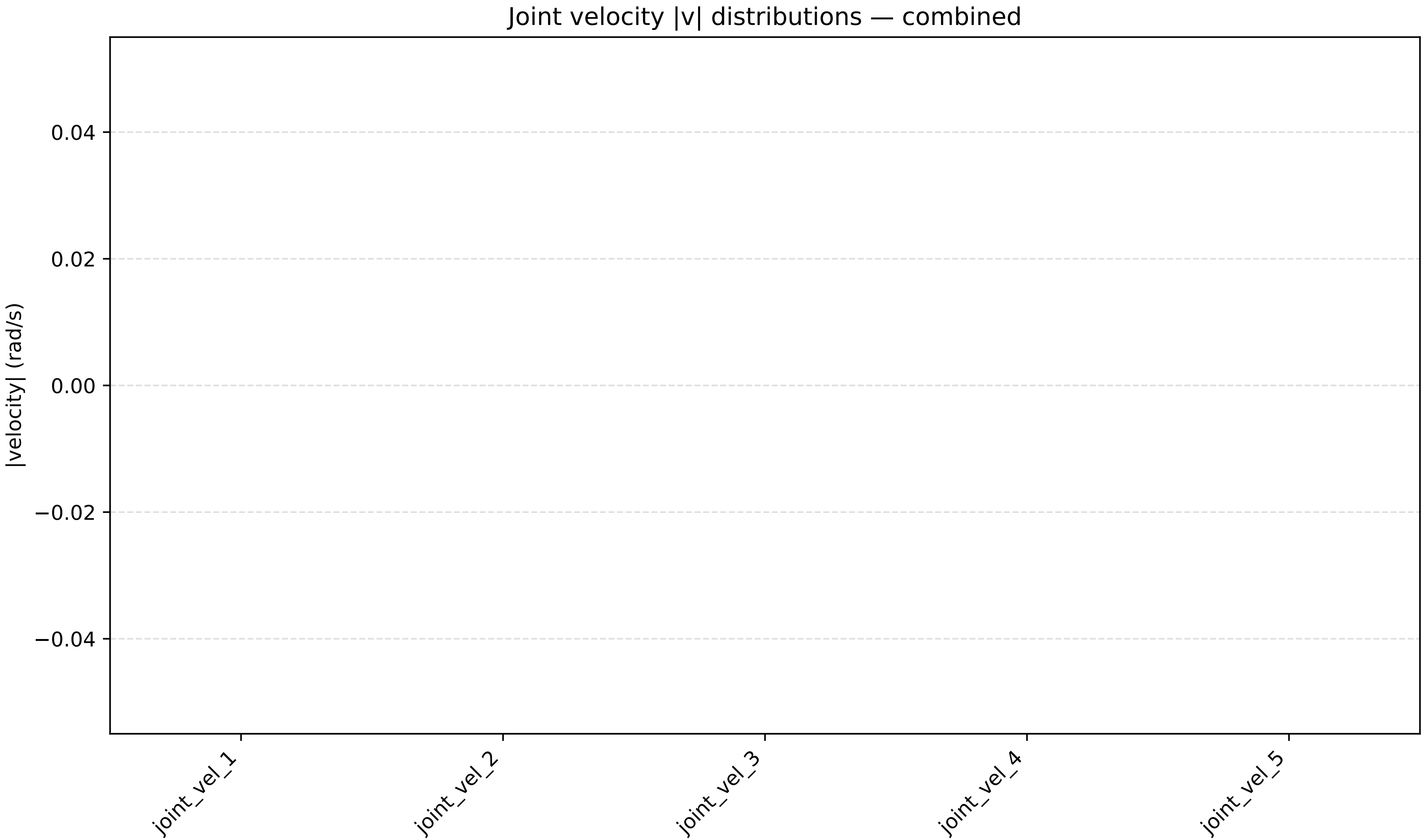
run_id	start_idx	end_idx	n_samples	duration_sec	success	pos_err	goal_x	goal_y	final_x	final_y	noise_strength_max	blur_strength_ma
970	970304	971303	1000	9.99	False	0.212750	0.2	-0.15	0.244220	0.062750	0.904662	0.00000
971	971304	972303	1000	9.99	True	0.014720	0.2	-0.15	0.213599	-0.164720	0.000000	0.00000
972	972304	973303	1000	9.99	True	0.018633	0.2	-0.15	0.202802	-0.131367	0.000000	0.00000
973	973304	974303	1000	9.99	False	0.062445	0.2	-0.15	0.259720	-0.087555	0.000000	0.80345
974	974304	975303	1000	9.99	False	0.202012	0.2	-0.15	0.264643	0.052012	0.000000	0.00000
975	975304	976303	1000	9.99	False	0.024560	0.2	-0.15	0.198691	-0.174560	0.000000	0.01989
976	976304	977303	1000	9.99	False	0.069751	0.2	-0.15	0.193132	-0.080249	0.000000	0.00000
977	977304	978303	1000	9.99	False	0.159975	0.2	-0.15	0.277592	0.009975	0.000000	0.00000
978	978304	979303	1000	9.99	False	0.276654	0.2	-0.15	0.209885	0.126654	0.000000	0.92019
979	979304	980303	1000	9.99	True	0.002080	0.2	-0.15	0.202080	-0.148788	0.000000	0.30444
980	980304	981303	1000	9.99	True	0.011833	0.2	-0.15	0.211833	-0.150913	0.000000	0.15101
981	981304	982303	1000	9.99	False	0.243838	0.2	-0.15	0.280455	0.093838	0.590819	0.00000
982	982304	983303	1000	9.99	False	0.181786	0.2	-0.15	0.241029	0.031786	0.379273	0.00000
983	983304	984303	1000	9.99	False	0.294887	0.2	-0.15	0.284675	0.144887	0.000000	0.92948
984	984304	985303	1000	9.99	True	0.018172	0.2	-0.15	0.218172	-0.149988	0.000000	0.00000
985	985304	986303	1000	9.99	False	0.299881	0.2	-0.15	0.273881	0.149881	0.000000	0.00000
986	986304	987303	1000	9.99	False	0.243763	0.2	-0.15	0.290232	0.093763	0.000000	0.97190
987	987304	988303	1000	9.99	True	0.008837	0.2	-0.15	0.208837	-0.148516	0.023161	0.00000
988	988304	989303	1000	9.99	False	0.214625	0.2	-0.15	0.279118	0.064625	0.744035	0.00000
989	989304	990303	1000	9.99	True	0.013347	0.2	-0.15	0.213347	-0.155881	0.000000	0.00000
990	990304	991303	1000	9.99	True	0.009554	0.2	-0.15	0.204219	-0.140446	0.000000	0.00000
991	991304	992303	1000	9.99	True	0.013351	0.2	-0.15	0.213303	-0.163351	0.000000	0.00000
992	992304	993303	1000	9.99	False	0.203627	0.2	-0.15	0.247514	0.053627	0.000000	0.00000
993	993304	994303	1000	9.99	True	0.013557	0.2	-0.15	0.186443	-0.142419	0.000000	0.00000
994	994304	995303	1000	9.99	False	0.056245	0.2	-0.15	0.256245	-0.109567	0.000000	0.00000
995	995304	996303	1000	9.99	False	0.219091	0.2	-0.15	0.253563	0.069091	0.000000	0.63807
996	996304	997303	1000	9.99	False	0.065679	0.2	-0.15	0.265679	-0.126540	0.549505	0.00000
997	997304	998303	1000	9.99	True	0.013983	0.2	-0.15	0.194167	-0.163983	0.002616	0.00000
998	998304	999303	1000	9.99	True	0.012497	0.2	-0.15	0.205600	-0.137503	0.000000	0.03819
999	999304	1000303	1000	9.99	True	0.008961	0.2	-0.15	0.208961	-0.146637	0.000000	0.00000

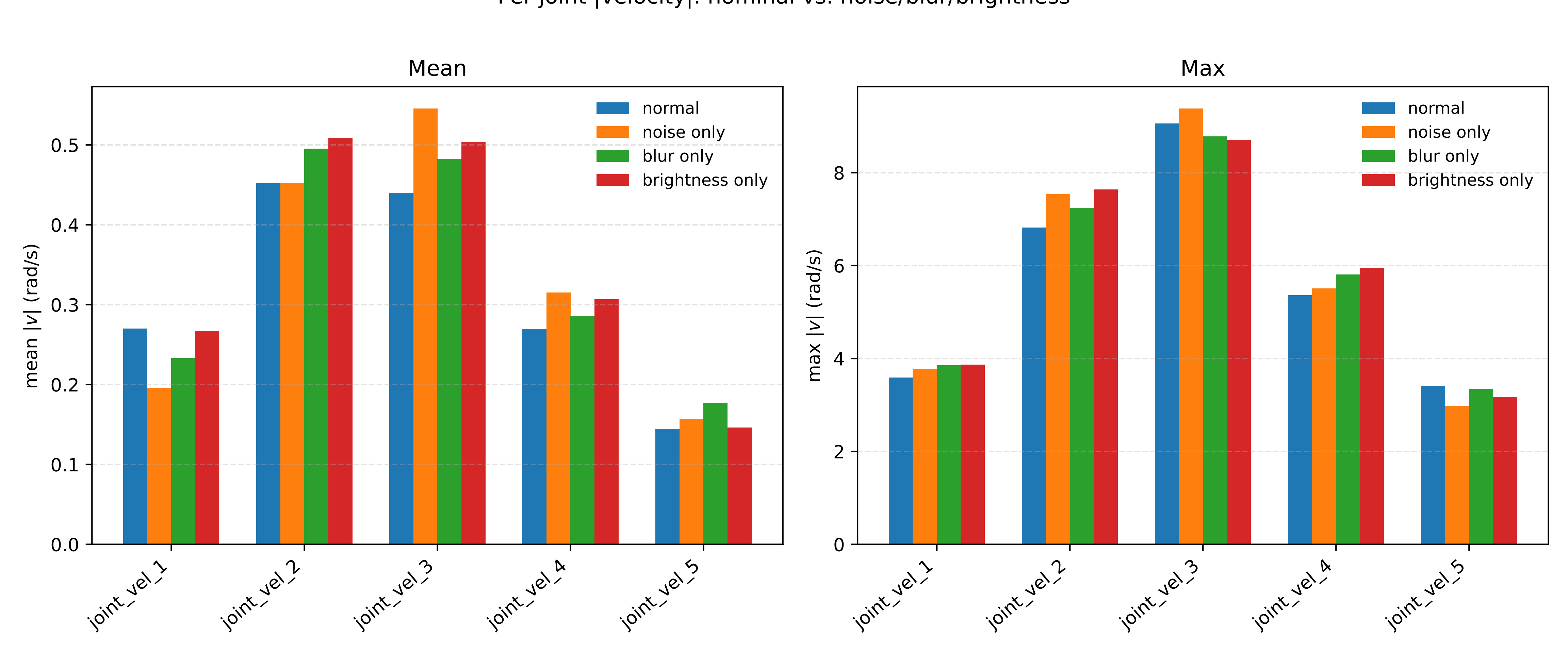

Joint velocity |v| distributions — normal
|velocity| (rad/s)
0.0
0.2
0.4
0.6
0.8
1.0
joint_vel_1
joint_vel_2
joint_vel_3
joint_vel_4
joint_vel_5


Joint velocity |v| distributions — noise_only
|velocity| (rad/s)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
joint_vel_1
joint_vel_2
joint_vel_3
joint_vel_4
joint_vel_5


Joint velocity |v| distributions — blur_only
|velocity| (rad/s)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
joint_vel_1
joint_vel_2
joint_vel_3
joint_vel_4
joint_vel_5


Joint velocity |v| distributions — brightness_only
|velocity| (rad/s)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
joint_vel_1
joint_vel_2
joint_vel_3
joint_vel_4
joint_vel_5


Joint velocity |v| distributions — combined
|velocity| (rad/s)
0.04
0.02
0.00
−0.02
−0.04
joint_vel_1
joint_vel_2
joint_vel_3
joint_vel_4
joint_vel_5


Per-joint |velocity|: nominal vs. noise/blur/brightness
Mean
Max
normal
noise only
blur only
brightness only
mean |v| (rad/s)
max |v| (rad/s)
0.0
0.1
0.2
0.3
0.4
0.5
0
2
4
6
8
joint_vel_1
joint_vel_2
joint_vel_3
joint_vel_4
joint_vel_5